Seb Insua *Software Engineer*

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Summary

Product engineer who builds real-time trading UIs end to end, from rendering and data to delivery, auth and operations. I ship major features into live, business-critical systems by designing for expressiveness, invariants, and additive change.

Operating Principles: end-to-end product development and architecture; 12-factor back-ends; low-latency UIs; strongly-typed data; tooling; inlining and colocation; defactoring; vertical integration; working in public; writing cultures; rejecting the problem.

Languages: TypeScript/JavaScript, Go, C#, Python, Rust, SQL, Shell/Bash, HTML, CSS.

Education: University of Kent, Bachelor's degree in Computer Science (2005 – 2009).

Experience

Maven Securities

Developer *TypeScript, React, Node.js, Go, C#/.NET, Postgres, SQL, Prometheus/Grafana* *Jan 2024 – Present*

- Overhauled high-throughput canvas grid for low-latency trading UI, adding zoom, hit-testing, SVG rasterization, high-DPI, seamless recovery from GPU driver crashes, and easier column resizing/reordering via dynamic hitboxes.
- Built the multi-version webserver and control plane for the trading UI; per-user channel-to-version resolution enabled staged rollouts and instant rollback, with every released version permanently servable.
- Built a production-safe observability and state-forensics CLI for the main trading application, spanning Postgres, logs and metrics; used to reconstruct user workspaces, inspect deploy state, and triage live issues.
- Diagnosed and resolved long-lived production-client reliability issues, including a network-layer memory leak (`AbortSignal + ky`) and a Chrome speech-synthesis failure that caused traders to lose audio for hours.
- Designed adaptive fuzzy matcher (two-pointer w/ contiguous-match bias + greedy fallback); supported multi-part queries, ranked matches via custom heuristic scoring and CSS-highlighted results.
- Implemented OIDC auth proxy in Go; supported silent refresh, cross-origin embedding, session resilience and complex flows; deployed in front of the production trading UI.
- Designed and delivered a recurring 5-day UI engineering training program for interns and graduates.

Freelance Engineer

Freelance & Private Study *Python, Rust, TypeScript* *Apr 2023 – Jan 2024*

- Adaptively stabilized streaming LLM output using exponential smoothing over queue growth to reduce jitter.
- Built a CLI for managing LLM app configs via declarative YAML; supported interactive prompts, workspace sync, and deployment flows.
- Wrote technical articles on algorithms, linear algebra, and how neural networks learn; implemented a neural network in Rust (micrograd-rs).

Software Engineer (Contract) at T. Rowe Price *TypeScript, Python, React, Node.js* *Oct 2020 – Mar 2023*

- Built synthetic monitoring service using Playwright; evolved from ASCII snapshot testing to DataFrame-style invariant checks (null detection, bounds, column relationships, sum validation) running against production.
- Rewrote legacy React application into TypeScript; removed Redux-as-cache layer, doubled Lighthouse score (50 → 98), and significantly improved grid testability.
- Improved reliability of Python, Node, and Kotlin services for data ingestion into S3/PostgreSQL; fixed major bugs in shared libraries for logging, tracing, and scheduling.

Software Architect (Contract) at JPMorgan Chase & Co. *TypeScript, React, Argo, Kubernetes* *Aug 2019 – Oct 2020*

- Re-hired into newly formed Core UI Infrastructure team to build tooling for high-scale development. Developed CI caching infra and Argo workflows that cut E2E test time from hours to minutes.
- Led upgrade of 50+ component library from Material UI v3 → v4 using codemods, visual diffing, and type-checking.
- Contributed to monorepo strategy and shipped DX tools forming the org's "golden path" for frontend dev.

Technical Lead (Contract) at Shell *Python, Jupyter, Spark, Kubernetes* *Dec 2018 – Jun 2019*

- Brought in by a former colleague to take over a politically sensitive project; aligned global stakeholders, hired a new team, and replaced consultancy build with in-house platform.

Application Engineer (Contract) at JPMorgan Chase & Co. *TypeScript, React* *Jun 2017 – Dec 2018*

- Led core UI work on the rewrite of 'Execute', JPMorgan's single-dealer trading platform, migrating from Flash to a modular React/TypeScript system under a hard end-of-life deadline.
- Established project governance: created RFC process, authored proposals (entitlements, module loading, CI), and aligned multiple teams across a shared monorepo (200+ packages).
- Built shell and widget tiling system; engineered internal toolchain for ESM/CommonJS/types using Rollup, Babel, and TypeScript. Created Danger plugin to deploy apps/storybooks for interactive PR review.

Earlier

Contract, lead and founding roles *Oct 2010 – Apr 2017*

- YLD (consultant), McKinsey & Company (external advisor), Home Office (tech lead), We R Interactive (lead game developer), Saffron Digital (lead Python developer; founding back-end engineer on HTC Watch through HTC's $48m acquisition); client work for Hailo, Keen.IO and the Economist.